

On this page

Configuration Recommendations



For the full Pulse configuration refer to the official [Configuration Reference](#) and the recommended [Production Setup](#).

Setting up the JVM DNS cache TTL

A couple of security properties should be set up. This can be achieved by:

1. Making sure that the `security.overridePropertiesFile` property is set to true in the `$JAVA_HOME/jre/lib/security/java.security` file.
2. Creating a new file (e.g. `/opt/feedzai/pulse/conf/java.security.properties`) with the following content :

```
# conf/java.security.properties
networkaddress.cache.ttl=1
networkaddress.cache.negative.ttl=3
```

Copy

1. Using the previously created file to set up the security properties on Pulse by adding `-Djava.security.properties=conf/java.security.properties` to the `JVM_OPTS` variable on `/opt/feedzai/pulse/bin/startup.sh` :

```
# /opt/feedzai/pulse/bin/startup.sh snippet
JVM_OPTS="( ... ) -Djava.security.properties=conf/java.security.properties"
```

Copy

1. Using the previously created file to set up the security properties on App engines by adding `-Djava.security.properties=conf/java.security.properties` to the `pulse.global.appengine.jvmopts` property on Pulse properties.

```
# /opt/feedzai/pulse/conf/pulse.properties snippet
pulse.global.appengine.jvmopts="( ... ) -Djava.security.properties=conf/java.security.properties"
```

Copy



Setting up these properties affects the cache behavior of all network address name resolutions for both Pulse server and App engine processes.

From the `java.security` file, next to the `networkaddress.cache.ttl` property on, for example, Zulu JDK:

"NOTE: setting this to anything other than the default value can have serious security implications. Do not set it unless you are sure you are not exposed to DNS spoofing attack."

Ansible



Available ansible variables for running the ansible playbook.

Variable	Default Value
ansible_user	feedzai
ansible_local_user	{{ ansible_user }}
pulse_install_path	"/opt/feedzai"
pulse_home_path	"{{ pulse_install_path }}/product-pulse-{{ pulse_version }}"
pulse_license_src_path	{{ basesoftware_path }}/pulse-licenses
pulse_sso_idp_cert	N/A
pulse_sso_private_key	N/A
pulse_sso_public_key	N/A
pulse_sso_force_https	false
pulse_api_url	"{{ jwk_service_host }}"
cm_sso_private_key	N/A
cm_sso_public_key	N/A
reference_data_servers	reference-data
zookeeper_nodes	zookeeper
cassandra_nodes	cassandra

Variable	Default Value
messaging_brokers	"rabbitmq:5672"
keycloak_url	"http://{{ keycloak_hostname }}:{{ keycloak_port }}"
keycloak_hostname	keycloak
keycloak_port	8080
cassandra_pulse_keyspace_replication	"class:'NetworkTopologyStrategy', 'dc1': 1"
cassandra_cluster_name	"{{ inventory }} cluster"
cassandra_pulse_keyspace	"profiles"
cassandra_table_name	"profiles"
database_pulse_eventstorage_hostname	"postgres"
database_pulse_eventstorage_port	"5432"
database_pulse_eventstorage_database_name	"feedzai"
database_pulse_metadata_hostname	"postgres"
database_pulse_metadata_port	"5432"
database_pulse_metadata_database_name	"feedzai"
database_pulse_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_eventstorage_read_only_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_port }}"

Variable	Default Value
database_pulse_metadata_jdbc	"jdbc:postgresql://{{ database_pulse_metadata_hostname }}:{{ database_pulse_metadata_port }}/{{ database_pulse_metadata_database_name }}?currentSchema=pulse&cancelSignalTimeout=2&connectTimeout=2&loadBalance=true"
database_pulse_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_eventstorage_hostname }}:{{ database_pulse_eventstorage_port }}/{{ database_pulse_eventstorage_database_name }}?currentSchema=eventstorage&cancelSignalTimeout=2&connectTimeout=2&loadBalance=true"
database_pulse_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_eventstorage_read_only_hostname }}:{{ database_pulse_eventstorage_read_only_port }}/{{ database_pulse_eventstorage_read_only_database_name }}?currentSchema=eventstorage&cancelSignalTimeout=2&connectTimeout=2&loadBalance=true"
database_pulse_outbound_payments_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_outbound_payments_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_outbound_payments_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_outbound_payments_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_outbound_payments_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_outbound_payments_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"
database_pulse_outbound_payments_eventstorage_username	"{{ database_pulse_eventstorage_username }}"
database_pulse_outbound_payments_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_outbound_payments_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"

Variable	Default Value
database_pulse_outbound_payments_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_outbound_payments_eventstorage_{{database_pulse_outbound_payments_eventstorage_database_name }}}}
database_pulse_outbound_payments_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_outbound_payments_eventstorage_{{database_pulse_outbound_payments_eventstorage_read_only_database_name }}}}
database_pulse_inbound_payments_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_inbound_payments_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_inbound_payments_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_inbound_payments_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_inbound_payments_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_inbound_payments_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"
database_pulse_inbound_payments_eventstorage_username	"{{ database_pulse_eventstorage_username }}"

Variable	Default Value
database_pulse_inbound_payments_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_inbound_payments_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"
database_pulse_inbound_payments_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_inbound_payments_eventstorage_h {{database_pulse_inbound_payments_eventstorage_database_name }}?"
database_pulse_inbound_payments_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_inbound_payments_eventstorage_re {{database_pulse_inbound_payments_eventstorage_read_only_database
database_pulse_cards_eventstorage_hostname	"{{ database_pulse_eventstorage_hostname }}"
database_pulse_cards_eventstorage_port	"{{ database_pulse_eventstorage_port }}"
database_pulse_cards_eventstorage_database_name	"{{ database_pulse_eventstorage_database_name }}"
database_pulse_cards_eventstorage_read_only_hostname	"{{ database_pulse_eventstorage_read_only_hostname }}"
database_pulse_cards_eventstorage_read_only_port	"{{ database_pulse_eventstorage_read_only_port }}"
database_pulse_cards_eventstorage_read_only_database_name	"{{ database_pulse_eventstorage_read_only_database_name }}"

Variable	Default Value
database_pulse_cards_eventstorage_username	"{{ database_pulse_eventstorage_username }}"
database_pulse_cards_eventstorage_password	"{{ database_pulse_eventstorage_password }}"
database_pulse_cards_eventstorage_schema	"{{ database_pulse_eventstorage_schema }}"
database_pulse_cards_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_cards_eventstorage_hostname }}:{{ database_pulse_cards_eventstorage_jdbc_opts join('&') }}"
database_pulse_cards_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_cards_eventstorage_read_only_hostname }}:{{ database_pulse_cards_eventstorage_jdbc_opts_read_only join('&') }}"
database_pulse_reference_data_eventstorage_hostname	"{{ database_pulse_outbound_payments_eventstorage_hostname }}"
database_pulse_reference_data_eventstorage_port	"{{ database_pulse_outbound_payments_eventstorage_port }}"
database_pulse_reference_data_eventstorage_database_name	"{{ database_pulse_outbound_payments_eventstorage_database_name }}"
database_pulse_reference_data_eventstorage_read_only_hostname	"{{ database_pulse_outbound_payments_eventstorage_read_only_hostname }}"
database_pulse_reference_data_eventstorage_read_only_port	"{{ database_pulse_outbound_payments_eventstorage_read_only_port }}"
database_pulse_reference_data_eventstorage_read_only_database_name	"{{ database_pulse_outbound_payments_eventstorage_read_only_database_name }}"
database_pulse_reference_data_eventstorage_username	"{{ database_pulse_outbound_payments_eventstorage_username }}"

Variable	Default Value
database_pulse_reference_data_eventstorage_password	"{{ database_pulse_outbound_payments_eventstorage_password }}"
database_pulse_reference_data_eventstorage_schema	"{{ database_pulse_outbound_payments_eventstorage_schema }}"
database_pulse_reference_data_eventstorage_jdbc	"jdbc:postgresql://{{ database_pulse_reference_data_eventstorage_host }}:{{ database_pulse_reference_data_eventstorage_jdbc_opts join('&') }}"
database_pulse_reference_data_eventstorage_jdbc_read_only	"jdbc:postgresql://{{ database_pulse_reference_data_eventstorage_read_only_host }}:{{ database_pulse_reference_data_eventstorage_read_only_jdbc_opts join('&') }}"
jwk_service_db_jdbc	jdbc:postgresql://postgres:5432/feedzai?currentSchema=jwkservice
jwk_service_db_schema	jwkservice
jwk_service_db_username	jwkservice
jwk_service_db_password	secret
jwk_service_host	"http://localhost:8080"
rabbitmq_brokers	"{{ messaging_brokers }}"
rabbitmq_pulse_username	"guest"
rabbitmq_pulse_password	"guest"
rabbitmq_pulse_vhost	"/"

Variable	Default Value
rabbitmq_feedback_storage_service_vhost	"feedback-storage"
rabbitmq_casemanager_brokers	"{{ rabbitmq_brokers }}"
rabbitmq_casemanager_username	"guest"
rabbitmq_casemanager_password	"guest"
rabbitmq_casemanager_vhost	"/"
database_pulse_metadata_username	"pulse"
database_pulse_metadata_password	"secret"
database_pulse_metadata_schema	"pulse"
database_pulse_eventstorage_username	"eventstorage"
database_pulse_eventstorage_password	"secret"
database_pulse_eventstorage_schema	"eventstorage"
database_pulse_app_metadata_username	"{{ database_pulse_metadata_username }}"

Variable	Default Value
database_pulse_app_metadata_password	"{{ database_pulse_metadata_password }}"
database_pulse_app_metadata_schema	"{{ database_pulse_metadata_schema }}"
pull_hadoop_files_from_bucket	false
hadoop_files_download_dir	"/tmp"
gcp_bucket_name	"pulse-dataproc-config"
gcp_hadoop_conf_file	"hadoop-files.tar.gz"
model_training_directory	"hdfs:///user/pulse/models"
model_storage_directory	"hdfs:///user/pulse/models"
model_hpt_directory	"hdfs:///user/pulse/hpt-models"
pulse_liquibase_environment	

Variable	Default Value
pulse_commands_environment	
pulse_setup_thread_dumps_cron	false
pulse_thread_dumps_cron_jobs	[]
number_threads_workflow_recovery	1
import_tmx_udf	true
import_fiv_markers_udf	true
cross_channel_metrics_enabled	false
cross_channel_metrics_rmi_based_lookup	true
cross_channel_metrics_fetch_timeout_ms	200
cross_channel_metrics_num_threads	1
cross_channel_metrics_client_consumer_threads	8

Variable	Default Value
cross_channel_metrics_server_consumer_threads	8
cross_channel_metrics_require_workflow_ready	true
cross_channel_metrics_publisher_pool_size	32
cross_channel_metrics_publisher_queue_size	1000
cross_channel_metrics_timed_futures_pool_size	100

Variable	Default Value
cross_channel_metrics_message_ttl_ms	1000
cross_channel_metrics_publish_timeout_ms	1000
cross_channel_metrics_batched_requests_enabled	false
cross_channel_metrics_batch_max_size	10
cross_channel_metrics_batch_max_time_ms	10
cross_channel_metrics_max_inflight_size	500
reference_data_tenant_strategy	STATIC

Variable	Default Value
inbound_payments_tenant_strategy	STATIC
transfers_tenant_strategy	STATIC
digital_activity_data_tenant_strategy	STATIC
cards_tenant_strategy	STATIC
inbound_payments_tenant_api_payload_field	customer_portfolio_country
outbound_payments_tenant_api_payload_field	customer_portfolio_country
digital_activity_tenant_api_payload_field	customer_portfolio_country
cards_tenant_api_payload_field	customer_portfolio_country

Variable	Default Value
reference_data_tenant_api_payload_field	customer_country_code
reference_data_tenant_provider_parameter_key	tenant
reference_data_tenant_provider_fallback	feedzai
pulse_spark_eventlog_enabled	false
pulse_spark_eventlog_dir	"hdfs:///var/log/spark/apps"
tars_enabled	false
node_app_pool	"{{ lookup('ansible.builtin.env', 'NODE_APP_POOL') }}"
pulse_app_topology	
feature_plan_log_level	Info

Variable	Default Value
watchdog_check_period_millis	10000
watchdog_latency_actuation_cycles	5
cards_customer_enrichment_key	customer_id
cards_account_enrichment_key	account_id
cards_card_enrichment_key	card_pan
cards_token_enrichment_key	ref_digital_wallet_token
numberOfPartitions	1

Variable	Default Value
cm_list_enrichment_enabled	true
cm_list_enrichment_prefix	"cmd_ "
additional_fields_enricher_active	true
additional_fields_enricher_fields	"pib_suspended, internal_payment_type, internal_event_source, isJade"
outbound_payments_feedback_strategy	SINGLE
inbound_payments_feedback_strategy	SINGLE
cards_feedback_strategy	SINGLE
partitioned_realtime_metrics_enabled	false

Variable	Default Value
rds_max_parallel_calls	4
workflow_validation_log_level	DEBUG
pulse_loggers	{}
lists_use_legacy_csv_format	false

Variable	Default Value
spark_datascience_jdk_home	
spark_distributed_jdk_home	
use_rsync	true
minimize_shuffles_enabled	true

Variable	Default Value
api_port	Outbound Payments: 8081 Inbound Payments:8084 Cards:8085
rds_api_port	8083
inbound_slack_future	1800000
inbound_slack_past	1800000
cards_slack_future	1800000
cards_slack_past	1800000

Variable	Default Value
transfers_slack_future	1800000
transfers_slack_past	1800000
digital_activity_slack_future	1800000
digital_activity_slack_past	1800000

Variable	Default Value
cards_offset_days	{debit: 4 credit: 5}
rabbitmq_queue_type	CLASSIC

(*) The configuration files should stored on a bucket in tar.gz format. The list of files inside the tar.gz should be:

- core-site.xml
- hdfs-site.xml
- mapred-site.xml
- yarn-site.xml

The service account used to fetch the configuration files from the bucket should have read permissions for that bucket (`storage.buckets.get` and `storage.buckets.list`).

Cron type

Variable	type	Mandatory	Default Value	Description
name	STRING	X		Name of the cron job
hour	STRING		*	The cron expression for <code>hour</code>
minute	STRING		*	The cron expression for <code>minute</code>
day	STRING		*	The cron expression for <code>day</code>
weekday	STRING		*	The cron expression for <code>weekday</code>
month	STRING		*	The cron expression for <code>month</code>

TARS application topology

```
pulse_app_topology:
  <app 1 id as defined in apps_enabled>:
```

```

# should be the amount of instances across all pools (for apps without partitioned workflows)
replication_factor: <app_1_replication_factor>
app_pools:
  # dictionary of pools with the number of instances per pool
  <app_1_pool_name>:
    number_of_instances: <app_1_number_of_instances_in_pool>
  <app_2_id as defined in apps_enabled>:
    # should be the amount of instances across all pools (for apps without partitioned workflows)
    replication_factor: <app_2_replication_factor>
    app_pools:
      # dictionary of pools with the number of instances per pool
      <app_2_pool_1_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_1>
      <app_2_pool_2_name>:
        number_of_instances: <app_2_number_of_instances_in_pool_2>

```

Copy

Where:

- `pulse_app_topology` is a dictionary:
 - each key is the application key as defined in `apps_enabled` and contains:
 - `replication_factor` : the total number of application instances (the sum of `app_pools[*].number_of_instances`)
 - a dictionary (`app_pools`) to define in which pools an application is deployed:
 - the key is the pool name
 - `number_of_instances` : the number of applications to deploy in the `pool`

By default this variable is not configured. An example configuration would be:

```

# Define topology structure for deployment with:
# -> One Pulse node running in the outbound pool
# -> Outbound and Reference Data applications running in the outbound node
pulse_app_topology:
  uk_outbound_payments:
    replication_factor: 1
    app_pools:
      outbound:
        number_of_instances: 1
  uk_reference_data:
    replication_factor: 1
    app_pools:
      outbound:
        number_of_instances: 1

```

Copy